Oxygen-Vacancy Energetics and Dopant-Induced Holes in Pr- and Rb-Substituted Anatase TiO_2: A DFT+U Comparison

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Abstract

Oxygen vacancies (V_O) in oxide electron-transport layers cause trap-assisted recombination; aliovalent substitution is the usual lever for controlling them, yet trivalent rare-earth and monovalent alkali dopants have rarely been compared under one protocol. We compare Pr^{3+} and Rb^{+} substitution on the Ti site of anatase TiO_2 with spin-polarised PBE+U calculations (U = 3.5 eV on Ti-3d, 2×2×1 supercell) and a common oxygen reference, μ_O = ½E(O_2). The neutral V_O formation energy falls from +4.74 eV (lower bound) in pristine anatase to +1.29 eV in the Pr-substituted cell and −4.06 eV in the Rb-substituted cell; the 5.36 eV Pr–Rb contrast changes by 0.04 eV on a 2×2×2 k-mesh. The ordering follows the formal electron count, one hole per Pr^{3+} and three per Rb^{+}, so one vacancy over-compensates Pr and leaves one residual hole with Rb. That hole is an empty minority-spin O-2p state; a Löwdin analysis puts 0.59 μ_B of its moment on one oxygen bonded to Rb, and the Rb ion lies 1.27 Å off its ideal site, towards the vacancy. The Rb cells carry unpaired spin in every treatment tested; the +2-charged vacancy cell reaches a total magnetisation of 2.99 μ_B. With U on Ti-3d only the Pr cells converge to non-magnetic smeared solutions, but the vacancy-free Pr cell develops one unpaired spin when an O-2p Hubbard term is added, so its spin state is method dependent. The results are consistent with charge-compensation control of V_O energetics; the two-dopant design does not separate oxidation state from size and bonding.

Keywords: DFT+U; anatase TiO_2; oxygen vacancy; aliovalent substitution; hole localisation; electron-transport layer.

1. Introduction

Indoor photovoltaics are a leading candidate to replace disposable batteries in the growing population of low-power sensors [1]. Because indoor illumination is orders of magnitude weaker than sunlight, the photocurrent is small and the open-circuit voltage is correspondingly sensitive to non-radiative recombination at defects, and wide-gap absorbers such as $CsPbBr_3$ [2] are paired with n-type oxide electron-transport layers (ETLs), for example TiO_2 or SnO_2 [3]. The same oxides serve as electron-transport layers in organic light-emitting diodes [4]. In TiO_2 the oxygen vacancy (V_O) is the dominant native defect: microscopy, photoemission, transient spectroscopy and first-principles work agree that its two excess electrons can either delocalise into the conduction band or localise as Ti^{3+} centres, and that the localised configuration acts as a trap [5–14]. Controlling the equilibrium V_O population of an ETL is therefore a materials-design problem, and substitutional aliovalent doping is a widely used lever [15–18].

For trivalent dopants on the Ti^{4+} site the formal charge mismatch is -1 e per site. Lanthanide substitution in anatase has been studied with DFT+U for Ce, Pr, Eu and Gd, including Pr together with an oxygen vacancy [19], and for Er at the (101) surface together with oxygen or titanium vacancies [20], and with plain GGA for Pr in a 12-atom bulk cell [21]; for Ce-doped zirconium titanate both hole formation and oxygen-vacancy formation have been considered as compensation channels, with electron paramagnetic resonance pointing to vacancy compensation [22]. The picture is not universal: Iwaszuk and Nolan showed for trivalent dopants in bulk rutile that oxygen-vacancy compensation can compete with, and in some cases dominate, electronic compensation [17]. The coupling between a lanthanide dopant and an oxygen vacancy has been analysed for Ce-doped anatase [23] and Gd-doped CeO_2 [18], and Bahmanrokh et al. proposed a defect-equilibria formalism for Ce-doped anatase in which both channels appear [24]. Pr, which has also been studied as an adsorbate on the anatase (101) surface [25], is an attractive test case, here modelled as trivalent with a 4f-in-core dataset (Section 2.1), because its ionic radius (0.99 Å in six-fold coordination [26]) is far from that of Ti^{4+} (0.605 Å). For monovalent alkali dopants the mismatch is -3 e per site, and the available evidence points consistently to V_O formation as the compensation channel. Carey and Nolan found that Li, Na, K and Rb substitution in α-Cr_2O_3 strongly reduces the V_O formation energy, with the larger alkalis having the larger effect [16]. Hybrid-functional work on alkali-doped α-Bi_2O_3 reached a similar conclusion [27], and Tian et al. recently combined experiment and calculation to show that Rb doping, in synergy with lattice strain, promotes oxygen-vacancy formation in TiO_2 [15].

What is missing is a comparison of the two dopant families in the same anatase host with the same supercell, pseudopotentials, Hubbard treatment and oxygen reference; the Rb case of Tian et al., for example, is embedded in a strained photochromic system and is not paired with a trivalent dopant. Without such a comparison, differences between published trivalent and monovalent results cannot be separated from differences in host, protocol or reference state. The comparison is also a useful test of the Kröger–Vink bookkeeping that is routinely invoked for dopant selection [28]: a Pr^{3+} substitution should leave one hole and a Rb^{+} substitution three, so a single neutral V_O (two electrons) should over-compensate Pr and under-compensate Rb.

This work provides that bounded comparison. Using one PBE+U protocol we compute the neutral V_O formation energy in pristine, Pr-substituted and Rb-substituted 2×2×1 anatase supercells, characterise the spin state and electron count of each cell, and test

the sensitivity of the localisation picture to an oxygen-site Hubbard term. We also report charged-cell spin trends, Bader-charge differences, a hybrid-functional check on the pristine cell, dispersion-corrected single points, a k-mesh verification and alternative vacancy sites. We state explicitly which conclusions the two-dopant design supports: it establishes the ordering and magnitude of the V_O energetics within this protocol and shows that they follow the formal charge count, but it does not isolate oxidation state from ionic radius or dopant-specific bonding, and it does not establish device-level recombination behaviour. The bulk formation energies reported here are the reference against which surface and interface V_O energetics in ETL films would be measured; interface models are outside the present scope. Section 2 describes the protocol, Section 3 reports structure, energetics, electronic structure and sensitivity tests, Section 4 discusses the results against prior work, Section 5 states the limitations and Section 6 concludes.

2. Computational Methods

2.1 Code, functional and pseudopotentials. All calculations were performed with Quantum ESPRESSO v7.5 [29,30] (hardware and wall times in Table S6) using the PBE exchange–correlation functional [31] and projector-augmented-wave (PAW) datasets from PSlibrary 1.0.0 [32]: Ti.pbe-spn-kjpaw_psl.1.0.0, O.pbe-n-kjpaw_psl.1.0.0, Pr.pbe-spdn-kjpaw_psl.1.0.0 and Rb.pbe-spn-kjpaw_psl.1.0.0. The valence configurations and electron counts of these datasets are listed in Table 1 because they fix the electron count of every supercell. The Pr dataset belongs to the PSlibrary “spdn” lanthanide family, which is generated for the trivalent ion with the 4f electrons frozen in the core; its 11 valence electrons comprise the 5s, 5p, 5d and 6s/6p shells (the generation configuration uses fractional 6s/6p occupations). The Pr-4f states are therefore not variational in the present calculations, and no conclusion about 4f occupation or 4f hybridisation can be drawn from them (Section 5). The Bader electron totals of the Pr_{Ti} + V_O and Rb_{Ti} + V_O cells (376.998 and 374.998 e, Table S3) reproduce the counts in Table 1 and confirm the valence assignment. Table S7 lists the MD5 checksums printed by pw.x (Ti, O and Rb; the Pr checksum was not recorded) and the atomic wavefunctions that projwfc.x reads from each dataset (Ti 3s/3p/3d, O 2s/2p, Pr 5p/5d, Rb 4s); the Pr dataset contains no 4f wavefunction. Kinetic-energy cut-offs of 40 Ry (wavefunctions) and 320 Ry (charge density) were used throughout. No cut-off convergence test was performed; comparable anatase defect work uses 55 Ry [13], and the consequences are discussed in Section 5.

Table 1. Pseudopotential valence configurations and resulting supercell electron counts. *N_e is the number of valence electrons per cell; all counts are for neutral cells. “Formal carriers” is the Kröger–Vink bookkeeping for Pr^{3+}, Rb^{+}, Ti^{4+} and O^{2-} with two electrons per neutral V_O; it does not state where the carriers localise. The cell labels in the first column (e.g. Pr_VO for $PrTi_{15}O_{31}$) are used throughout the text.*

Species / cell	Valence configuration (PSlibrary 1.0.0)	Z_{val}	Composition	N_e	Formal carriers
Ti (spn)	$3s^2 3p^6 3d^2 4s^2$	12	–	–	–
O (n)	$2s^2 2p^4$	6	–	–	–
Pr (spdn, 4f in core)	$5s^2 5p^6 5d^1 (6s,6p)^2$	11	–	–	–
Rb (spn)	$4s^2 4p^6 5s^1$	9	–	–	–
Pristine_perfect	–	–	$Ti_{16}O_{32}$	384	none

Species / cell	Valence configuration (PSlibrary 1.0.0)	Z_{val}	Composition	N_e	Formal carriers
Pristine_VO	–	–	$Ti_{16}O_{31}$	378	2 excess electrons
Pr_perfect	–	–	$PrTi_{15}O_{32}$	383	1 hole
Pr_VO	–	–	$PrTi_{15}O_{31}$	377	1 net excess electron
Rb_perfect	–	–	$RbTi_{15}O_{32}$	381	3 holes
Rb_VO	–	–	$RbTi_{15}O_{31}$	375	1 residual hole

2.2 Hubbard correction. A Hubbard term in the rotationally invariant Dudarev form [33], which is the form implemented behind the HUBBARD card of Quantum ESPRESSO 7.x, was applied to the Ti-3d manifold with U_{eff} = 3.5 eV using ortho-atomic projectors; U_{eff} is written U, U(Ti-3d) or U(O-2p) in the rest of the paper. This value is close to the first-principles linear-response value for the Ti-3d subspace of anatase (U = 3.57 eV, U_{eff} = U − J = 3.28 eV, with LDA [34]), and it has been used for Er-doped anatase [20] and for anatase (101) defect chemistry [7]; other anatase work uses U = 4.2 eV [13], a value originally fitted for the rutile (110) surface [35], or larger values [14]. Literature U values obtained with other projector schemes are not strictly transferable to the ortho-atomic projectors used here, and the sensitivity of the present results to U is reported in Section 3.7. The DFT+U approach rests on the formulation of Anisimov et al. [36]; linear-response determination of U [37] was not performed for any cell in this work. For the oxygen-site tests of Section 3.4 an additional U_{eff} = 5.5 eV was placed on the O-2p manifold. This value is transferred from the Cr_2O_3 study of Carey and Nolan [16], where it was used together with a Cr-3d term to localise oxygen holes; it has not been derived for anatase and is treated here as a sensitivity parameter rather than a calibrated value; for comparison, the linear-response value for the O-2p subspace of anatase is U_{eff} = 7.66 eV with LDA [34], and U can alternatively be fixed by enforcing piecewise linearity of the energy on polaronic defect states [38]; no such calibration was attempted here.

2.3 Supercells. A 2×2×1 anatase supercell with a = b = 7.63632 Å and c = 9.77900 Å (volume 570.25 $Å^3$) was used, containing 48 atoms ($Ti_{16}O_{32}$) in the perfect cell and 47 after removal of one oxygen atom. Dopants were introduced by replacing one Ti atom with Pr or Rb, i.e. 6.25 % of cation sites; one V_O per cell corresponds to 3.1 % of oxygen sites or 1.75×10^{21} cm^{-3}. These concentrations are far above the dilute limit relevant to ETLs, and all absolute quantities below should be read as those of a concentrated periodic defect array. Lattice parameters were taken from a variable-cell relaxation of the pristine cell and held fixed for all defect cells. That relaxation was performed on the 12-atom conventional cell; its k-mesh, cut-off and residual stress were not recorded in the run log. The production V_O is a first-shell oxygen of the dopant site: in the relaxed Rb_VO coordinates the vacant position is an equatorial oxygen site 1.95 Å from the ideal cation position (Section 3.1, Table S8), and in Pr_VO the four lowest oxygen Löwdin populations are on O9, O11, O23 and O30, four of the five remaining first-shell oxygens of the dopant site, while Ti2 and Ti26, the two Ti bonded to the vacant site in the Rb cell, are among the three Ti with the highest 3d population, which is consistent with the same vacancy in both cells. Three alternative sites in the Pr cell are examined in Section 3.7. The Pr_VO input was not re-inspected for this revision and the relaxed Pr_VO geometry was not analysed; the Pr–O distance of ≈2.17 Å quoted in earlier drafts is undocumented and is not used (the Shannon-radius sum for six-fold Pr^{3+}–O^{2-} is 2.39 Å [26]).

2.4 Brillouin-zone sampling and electronic settings. Production relaxations used Γ-point sampling; in earlier anatase defect work a Γ-only grid was used only for the hybrid-functional calculations, with denser meshes otherwise [14]. Single-point total energies on the Γ-relaxed geometries were recomputed on a 2×2×2 Monkhorst–Pack mesh for all four doped cells, and the projected densities of states and Löwdin moments were obtained from non-self-consistent calculations on a 2×2×2 mesh (Figure 4; Supporting Information, Section S2); the k-mesh dependence of the formation-energy contrast is reported in Section 3.7. All calculations were spin-polarised (nspin = 2). Fractional occupations were treated with Gaussian smearing. The single-point inputs derived from the production runs (U sweep, Section 3.7) use degauss = 0.005 Ry, whereas the converged pristine-V_O relaxation (Section S1) and the self-consistent runs that regenerated the densities for the projected densities of states used 0.01 Ry (Table S10); because the latter reproduce the production Rb_VO Fermi energy to 0.1 meV, the production value (0.005 or 0.01 Ry) is not established; the production input files were not re-inspected for this revision. With smearing, the total energy printed by pw.x is the smearing free energy; the entropy term is small when few states lie near the Fermi level but has not been tabulated. The −TS terms were not extracted from the outputs. For orientation, with Gaussian smearing a single level at E_F contributes $\sigma/(2\sqrt{\pi})$ per spin channel to −TS, about 0.02 eV at 0.005 Ry and 0.04 eV at 0.01 Ry; the term grows with the number of levels within σ of E_F and would have to reach several tenths of an eV to affect the spin-state comparisons of Section 3.3. The electronic self-consistency threshold was 1×10^{-8} Ry. Symmetry was handled by the code's default detection (nosym was not set, and no magnetisation constraint, tot_magnetization or constrained_magnetization, appears in the run log or in any input template); the Rb_VO cell retained a mirror plane (two symmetry operations in the pw.x output), so hole configurations that break it, for example localisation on only one of a mirror-related pair of oxygens, were excluded by construction.

2.5 Magnetisation conventions and spin initialisation. Two quantities are reported throughout: the total (net) magnetisation $m = \int[\rho_\uparrow(r) - \rho_\downarrow(r)]\,dr$, which can be of either sign, and the absolute magnetisation $m_{abs} = \int|\rho_\uparrow(r) - \rho_\downarrow(r)|\,dr$. For a single unpaired electron, localised or delocalised, both are close to 1 μ_B, so neither quantity measures localisation; $m_{abs} > |m|$ indicates spin density of both signs. The run log records starting_magnetization = 0 on every species for the production relaxations, and the single-point inputs derived from them use the same setting; pw.x accepts explicitly set zero values with nspin = 2 and stops only when starting_magnetization is set for no species and no total magnetisation is imposed. From a spin-symmetric start a moment develops only if the symmetric solution is unstable and numerical noise (for example from randomised starting wavefunctions, which the single-point inputs use) breaks the symmetry, as happened for both Rb cells; the same start can leave a calculation in a non-magnetic solution when a magnetic one exists. The zero start is recorded in the run log and in the single-point templates derived from the production inputs. For this reason the Pr_VO cell was re-relaxed from a magnetic start with starting_magnetization = 0.3 on Ti and 0.5 on the dopant under a separate prefix, and the two total energies were compared (Section 3.3). These starting values are fractions of the valence charge in the Quantum ESPRESSO convention, not site moments in μ_B. The converged pristine-V_O relaxation started from 0.05 on Ti and O and ended non-magnetic (Section S1), and the self-consistent run preceding the Rb_VO projected densities of states started from 0.3 on Ti and reproduced the production Fermi energy (7.1329 eV). No magnetic start was tried for Pristine_perfect or, with U on Ti-3d only, for Pr_perfect. A wider search over site-

specific and antiparallel seeds was not performed, and the non-magnetic solutions are therefore described below as the lowest-energy solutions among those tested rather than as established ground states.

2.6 Geometry optimisation. Ionic positions were relaxed with BFGS until the largest force component fell below 0.005 Ry/Bohr (0.129 eV/Å). For the doped cells, charge mixing was damped (mixing_beta = 0.1) and the BFGS history shortened (bfgs_ndim = 3) to suppress oscillation about shallow minima; the pristine V_O cell was first attempted with mixing_beta = 0.3 (mixing_mode = local-TF) and converged only after the changes described in Section S1. The final maximum forces were 0.0026, 0.0042, 0.0037 and 0.0040 Ry/Bohr (0.067, 0.108, 0.095 and 0.103 eV/Å) for Pr_perfect, Pr_VO, Rb_perfect and Rb_VO. The production Pristine_perfect energy (−4275.01530 Ry) is not from a relaxed geometry: the run log records it as a single-point calculation on the positions of the variable-cell relaxation of the conventional cell, with a residual maximum force of 0.0225 Ry/Bohr (0.58 eV/Å) in the 2×2×1 cell at Γ; the logged energy in eV converts to the same Rydberg value, and a from-scratch single point on the same positions reproduces it to 10^{-5} Ry (Table S2a). The doped perfect cells were relaxed after their unrelaxed single points were found to carry forces of 0.43–0.66 Ry/Bohr; the pristine perfect cell was not. Relaxing it would lower E(Pristine_perfect) and raise the pristine formation energy of Table 2, and the two reductions measured from it, by the relaxation energy, which was not computed (Section 5); the pristine E_f is therefore a lower bound, whereas the doped-cell formation energies and their contrast do not involve this cell. The Pr_VO relaxation was stopped manually after the force fell below the threshold while the BFGS step oscillated; the energy change over the last iterations was below 1×10^{-9} Ry. forc_conv_thr was 0.005 Ry/Bohr for every relaxation; etot_conv_thr was not recorded (pw.x default 1×10^{-4} Ry unless set). This tolerance is looser than the 0.01–0.03 eV/Å commonly used for defect energetics, and its consequences are discussed in Section 5. The BFGS force histories are contained in the deposited output files.

2.7 Formation energies. The neutral V_O formation energy of host X was evaluated as

$E_f(V_O^0; X) = E(X \text{ with } V_O, q = 0) - E(X \text{ perfect}, q = 0) + \mu_O$, with $\mu_O = ½E(O_2) + \Delta\mu_O$ and $\Delta\mu_O = 0$,

where $E(O_2)$ is the total energy of an isolated O_2 molecule in a 12 Å cubic box computed with the same settings; it converged to the triplet ground state (m = 2.00 μ_B) and $½E(O_2)$ = −41.51621 Ry (−564.857 eV). This is the oxygen-rich electronic-energy reference; no finite-temperature gas-phase correction or competing-phase analysis was applied, and the values are conditional oxygen-removal energies from the already substituted cells rather than equilibrium vacancy concentrations [39]. The known overbinding of O_2 by PBE [14] shifts all absolute formation energies at this reference by a common amount and cancels in differences between hosts; no empirical correction was applied. Formation energies were evaluated from unrounded Rydberg totals and are reported to 0.01 eV; the full-precision ledger is Table S9. The comparative quantity $\Delta E_f = E_f(V_O; Pr) - E_f(V_O; Rb)$ is independent of the oxygen reference.

2.8 Additional calculations. (i) Charged cells: Pr_VO and Rb_VO were relaxed at q = +1 and +2 with a uniform compensating background; no image-charge or potential-alignment correction was applied, so these cells are used only for their spin state and not for transition levels. (ii) U sweep: single-point calculations on the Pristine_perfect, Pr_VO

and Rb_VO geometries of the production runs at U(Ti-3d) = 3.0, 4.0 and 4.5 eV. (iii) HSE06: one single-point calculation on the production Pristine_perfect geometry with exact-exchange fraction 0.25 and screening parameter 0.106 $Bohr^{-1}$ [40,41], Γ-only, 40 Ry, without the Hubbard term. The recorded settings are input_dft = 'hse', exx_fraction = 0.25 and screening_parameter = 0.106 $Bohr^{-1}$ (0.2 $Å^{-1}$; Krukau et al. recommend 0.11 $Bohr^{-1}$ [41]), with the same PAW datasets as the production runs; the run log records no other exact-exchange setting, so the q-grid, ecutfock and EXX threshold are taken to be the pw.x defaults. The hybrid total energy (−3926.67 Ry) is not comparable with the PBE+U total of the same cell (−4275.02 Ry): the hybrid run has no Hubbard term, and pw.x evaluates the exact-exchange energy of PAW datasets from the smooth pseudo-wavefunctions, so the two totals share no reference. Only the qualitative outcome of this run is used (Section 3.6). (iv) Dispersion: single-point energies for Pr_VO and Rb_VO with the Grimme D3 correction and Becke–Johnson damping [42,43], without three-body terms (D3 has been combined with DFT+U for pristine and defective anatase by Torres et al., who used PBEsol-D3 with U = 3 eV for the defective (001) surface [44]); the single points were started from scratch on the production geometries. (v) Alternative vacancy sites: three additional Pr_VO cells with the vacancy at oxygen sites 1, 7 and 17 of the orchestration input (not the pw.x atom numbers used elsewhere). (vi) Bader analysis of Pr_VO and Rb_VO from pp.x charge densities with the Henkelman grid algorithm [45,46]. The pp.x density is the valence pseudo-density on the FFT grid of the 40/320 Ry calculation (plot_num = 0, which does not include the PAW all-electron reconstruction); grid convergence of the Bader charges was not tested. (vii) Projected densities of states, Löwdin charges and Löwdin spin polarisations from projwfc.x, on 2×2×2 non-self-consistent runs (tetrahedron occupations; 260 bands for Pr_VO and 226 for Rb_VO) that start from a Γ-only self-consistent density regenerated on the final relaxation geometry; Gaussian broadening 0.01 Ry (0.136 eV), energy step 0.02 eV, spilling parameters 0.0030 (Pr_VO) and 0.0032 (Rb_VO) (Section S2). Löwdin populations depend on the projector basis and are used only to locate the spin density.

3. Results

3.1 Structural relaxation

All four doped supercells relaxed to maximum forces below the 0.005 Ry/Bohr threshold (Section 2.6). The pristine V_O cell, which required a modified mixing scheme to converge (Section S1), was stopped at 0.0056 Ry/Bohr, slightly above the threshold. Figure 1 shows the (010) atomic plane that contains the dopant site and the vacancy, at ideal positions and in the relaxed Rb_VO cell as read from its final coordinates (Table S8). The vacant site is an equatorial first-shell oxygen of the dopant site, 1.95 Å from the ideal cation position. In the relaxed cell the Rb ion has moved 1.27 Å from that position towards the vacancy and lies 0.87 Å from the vacant oxygen site; its four nearest oxygens are at 2.54 (two), 2.56 and 2.71 Å and the next two at 2.95 and 3.07 Å. The two Ti atoms that were bonded to the removed oxygen have moved away from it, to 2.36 and 2.38 Å from the vacant site (ideal Ti–O distances 2.03 and 1.95 Å). The large monovalent ion thus occupies part of the space vacated by the oxygen. The displacement is measured from the ideal anatase site, because the relaxed Rb_perfect coordinates were not available, and the shortest Rb–O distance (2.54 Å) is 0.38 Å below the Shannon-radius sum for six-fold Rb^{+}–O^{2-} (2.92 Å [26]), as for the short Pr–O distance noted in Section 2.3. The relaxed Pr_VO geometry was not analysed in the same way, because its output was not available for this revision. Because the residual forces of the doped cells (0.07–0.11 eV/Å) are

larger than the 0.01–0.03 eV/Å typically used for defect energetics, bond lengths quoted here are indicative, and the sensitivity of the energetics to tighter relaxation has not been tested (Section 5).

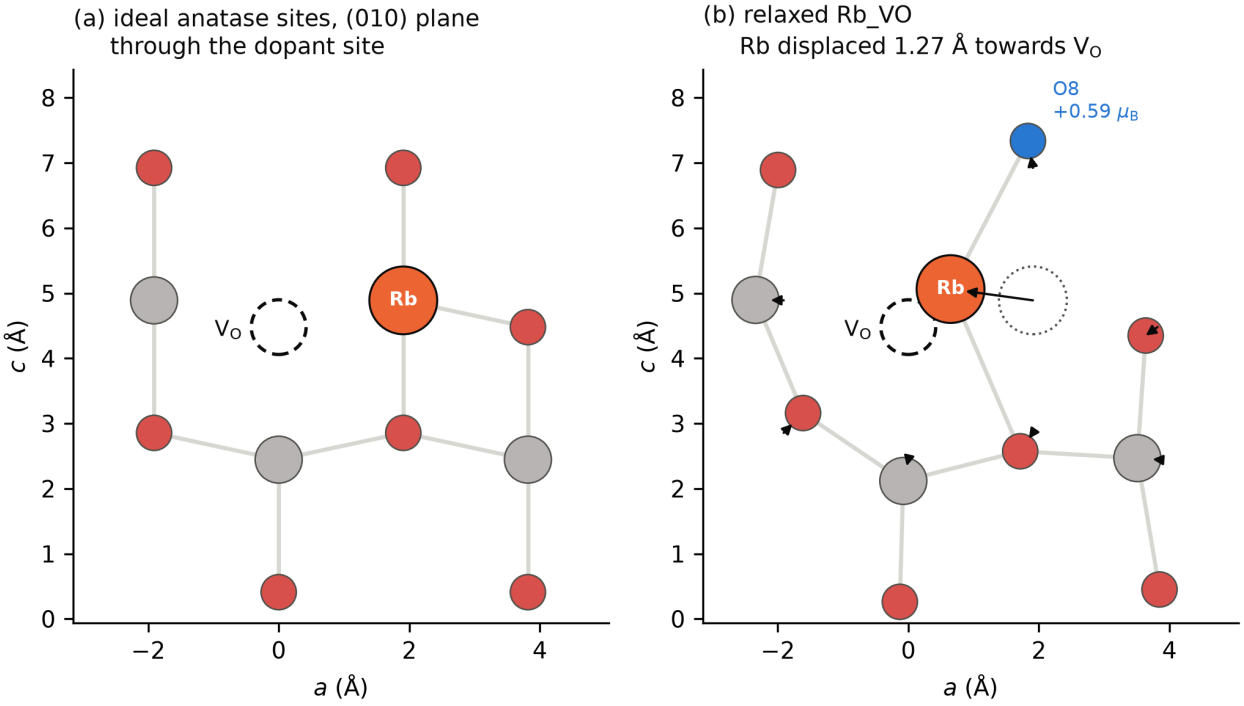


Figure 1. The (010) atomic plane containing the dopant site and the oxygen vacancy. (a) Ideal anatase positions built with the lattice parameters of this work (a = 7.63632 Å, c = 9.77900 Å, u = 0.208), with the dopant on the substituted Ti site and the vacant oxygen site marked by a dashed circle. (b) The same plane in the relaxed Rb_VO cell (PBE+U, U(Ti-3d) = 3.5 eV, Γ-only), from the final coordinates of the production relaxation. Arrows give the displacement (where larger than 0.2 Å) of each in-plane atom from its ideal site after removal of the mean displacement; the dotted circle marks the ideal Rb site. The oxygen drawn in blue (O8) carries the largest Löwdin spin polarisation (+0.59 μ_B; Section 3.6, Figure S1). Grey: Ti; red: O; orange: Rb. Only atoms in the plane are drawn, so each Ti shows fewer than six bonds. Plotted from the pw.x output by figures/make_fig1_structures.py.

3.2 Oxygen-vacancy formation energies

Table 2 lists the neutral V_O formation energies at the oxygen-rich reference $\mu_O = ½E(O_2)$, and Figure 2 displays them. In pristine anatase $E_f = +4.74$ eV, a lower bound because the perfect reference cell was not relaxed (Section 2.6). In the Pr-substituted cell $E_f = +1.29$ eV, and in the Rb-substituted cell $E_f = -4.06$ eV. Both substitutions lower the cost of removing an oxygen atom near the dopant site relative to pristine TiO_2, by at least 3.45 eV for Pr and 8.81 eV for Rb (differences of the unrounded values; lower bounds for the same reason). At this reference the Pr cell remains uphill, so oxygen removal is not spontaneous; the Rb cell is downhill, so the vacancy-free Rb-substituted cell is unstable against oxygen loss. Structural relaxation may contribute to the Rb reduction: the Rb ion lies 1.27 Å off its ideal site, in the space vacated by the oxygen (Section 3.1). The relaxation energy of the Rb_VO cell was not computed, and the pristine and Pr values

also contain relaxation, so its share of the reduction is unknown. The Pr–Rb contrast ΔE_f is 5.36 eV (5.356 eV from the unrounded totals; the value 5.35 eV obtained by subtracting the rounded entries is a rounding artefact).

Table 2. Neutral oxygen-vacancy formation energies at μ_O = ½E(O_2) (PBE+U, U(Ti-3d) = 3.5 eV, 40 Ry). *Total energies are the converged Γ-only values in Ry; E_f is computed from the unrounded totals (full ledger in Table S9). The last column gives the single-point result on the Γ-relaxed geometry with a 2×2×2 k-mesh. The Γ-to-2×2×2 shift is −0.10 eV (Pr) and −0.14 eV (Rb) and was not evaluated for the pristine formation energy; the Pr–Rb difference shifts by +0.04 eV. The 2×2×2 Pr_VO and Rb_VO energies are runs of the k-mesh series (Table S2a), which start from scratch and therefore include small offsets from the relaxation endpoints (Section 3.7). Smearing, cut-off and relaxation-tolerance contributions to the absolute values have not been quantified (Sections 2.4 and 5).*

Host X	E(X perfect) (Ry)	E(X + V_O) (Ry)	½E(O_2) (Ry)	$E_f(V_O^0)$, Γ (eV)	$E_f(V_O^0)$, 2×2×2 (eV)
Pristine TiO_2	−4275.01530 [b]	−4233.15036	−41.51621	+4.74	– [a]
Pr_{Ti}	−4576.71224	−4535.10106	−41.51621	+1.29	+1.19
Rb_{Ti}	−4330.38890	−4289.17135	−41.51621	−4.06	−4.20
ΔE_f (Pr − Rb)				+5.36	+5.39

[a] *The pristine V_O cell was not computed at 2×2×2 (the pristine perfect cell was, Section 3.7), so no 2×2×2 formation energy can be evaluated for the pristine host.* [b] *E(Pristine_perfect) is a single point on unrelaxed positions (F_{max} 0.0225 Ry/Bohr, Section 2.6), so the pristine E_f is a lower bound.*

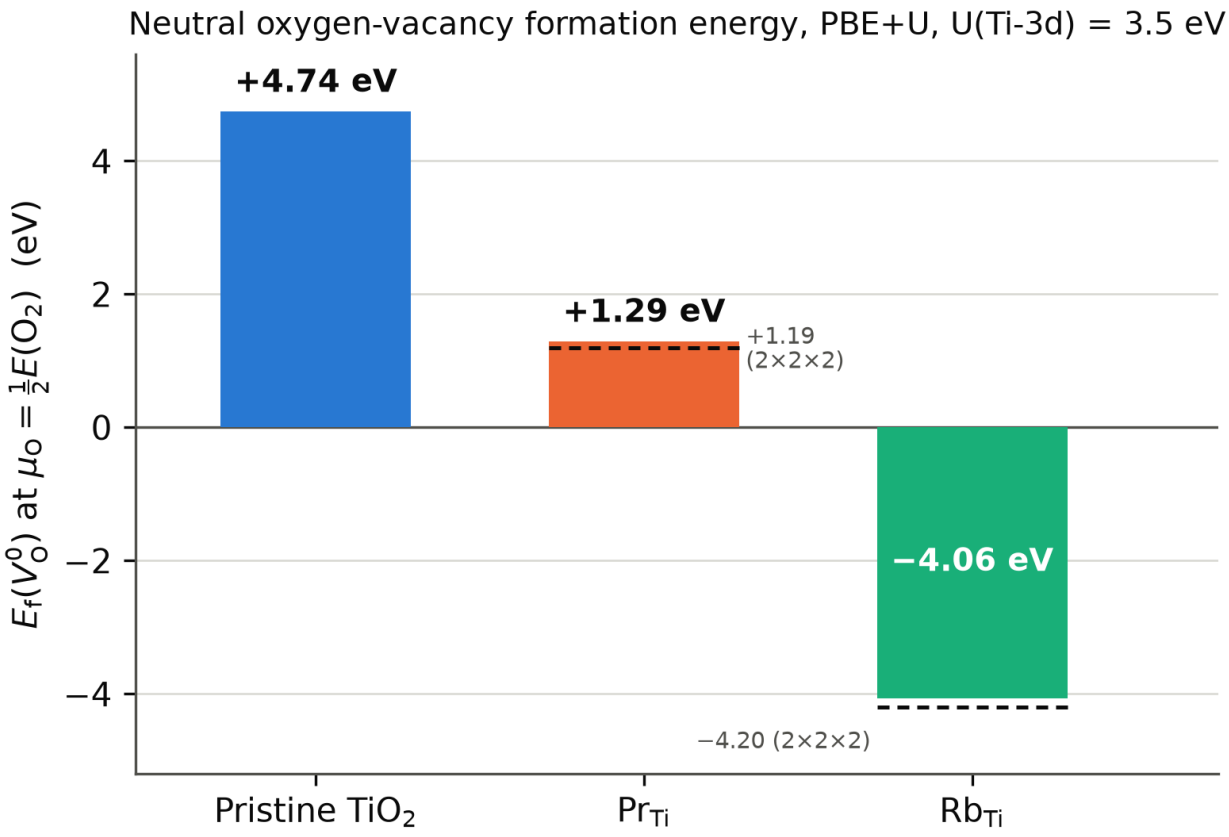


Figure 2. Neutral oxygen-vacancy formation energies at the oxygen-rich reference μ_O = ½E(O_2) for pristine, Pr-substituted and Rb-substituted anatase (PBE+U, U(Ti-3d) = 3.5 eV). Bars: Γ-only relaxed cells (this work; Table 2). Dashed ticks: single-point values at 2×2×2 on the Γ geometries. Both substitutions lower the formation energy relative to pristine; the Pr cell stays positive and the Rb cell becomes negative.

Previous bulk-anatase studies report the neutral-vacancy formation energy at the oxygen-rich limit only graphically, with PBE+U (U = 5.8 eV on Ti-3d) and a reduced-exchange hybrid (HSE15) by Arrigoni and Madsen [14] and with HSE06 in a 108-atom cell by Boonchun et al. [47]; because their protocols differ from the uncorrected PBE+U value at a PBE O_2 reference used here, a numerical comparison is not attempted, and the pristine value is used only as the internal reference for the two doped hosts. On the 2×2×2 mesh the individual formation energies shift by −0.10 eV (Pr) and −0.14 eV (Rb) while the contrast moves from 5.36 to 5.39 eV, i.e. by +0.04 eV from the unrounded totals (defined as the 2×2×2 contrast minus the Γ-only contrast). The comparative conclusion is therefore insensitive to Brillouin-zone sampling at the 0.05 eV level, although the absolute values carry a 0.10–0.14 eV mesh dependence. The direction of the Rb result (a reduction of E_f relative to pristine) agrees with the combined experimental and computational finding of Tian et al. that Rb doping promotes V_O formation in TiO_2 [15]; that work supports the direction of the effect, not the specific supercell value.

3.3 Spin states and electron counts

Table 3 collects the total and absolute magnetisation and the Fermi energy printed by the code for each production cell, and Figure 3 plots the magnetisations. Table 1 shows that every doped cell has an odd number of electrons. A collinear calculation of an odd-

electron cell can converge to m = 0 only through fractional occupations, so the non-magnetic solutions found for Pr_perfect and Pr_VO are smeared solutions in which the formal hole (Pr_perfect) or the formal net excess electron (Pr_VO) is carried by fractionally occupied states at E_F with equal spin-up and spin-down weight (conduction-band states in Pr_VO, Figure 4; not yet documented for Pr_perfect). They are not closed-shell singlets. The one magnetic trial performed for Pr_VO (Section 2.5) converged to m = 1.00 μ_B, m_{abs} = 1.13 μ_B. In smearing free energy (Section 2.4) it lies 0.281 eV above the non-magnetic energy logged with it for comparison and 0.394 eV above the final production energy (Table S9, runs 4, 24 and 25); this free energy includes a −TS term that favours fractionally occupied solutions. The non-magnetic solution is therefore the lower of the two solutions tested at this smearing, and a broader search over spin seeds and local distortions would be needed to establish it as the ground state. For Pr_VO the projected densities of states (Section 3.6) show that the net excess electron occupies Ti-3d conduction-band states with no gap state; because the non-self-consistent run inherits the m = 0 density, its equal spin channels and zero site moments follow by construction and do not test the spin state. The occupation numbers near E_F were not extracted from the outputs. The isolated O_2 reference converged to its triplet (m = 2.00 μ_B), which is the expected check of the spin-polarised pipeline, and the pristine V_O cell is non-magnetic (Section S1), in line with embedded coupled-cluster calculations that place the triplet of the neutral anatase vacancy 1.4 eV above a closed-shell colour-centre singlet [48].

The Rb cells behave differently. Rb_perfect converged to m = 1.00 μ_B (m_{abs} = 1.00 μ_B) from the production starting magnetisation, and Rb_VO to m = −0.97 μ_B with m_{abs} = 1.21 μ_B. Single-point calculations started from zero magnetisation on the final Rb_VO geometry, at the production U without and with the D3 term (Section 3.7) and at other U values (Table S1), converge instead to m = 1.00 μ_B. The two solutions are nearly degenerate: the from-scratch Γ single point (m = 1.00 μ_B, E_F = 7.120 eV) lies 6.4 meV above the last relaxation step (m = −0.97 μ_B, m_{abs} = 1.21 μ_B, E_F = 7.133 eV; Tables S2a and S9). The self-consistent run that preceded the projected densities of states, started from 0.3 on Ti, reproduced the production Fermi energy to 0.1 meV, but the two solutions differ in E_F by only 13 meV and that run's total energy was not recorded, so Figure 4 is taken to represent both near-degenerate solutions. With three formal holes in Rb_perfect and one residual hole in Rb_VO (Table 1), the Ti-3d-only Hubbard treatment thus yields one unpaired spin in each Rb cell; the remaining two holes of Rb_perfect are spin-paired in this treatment (m_{abs} = |m| excludes antiparallel moments), a point that the oxygen-site Hubbard test of Section 3.4 revisits. The excess of m_{abs} over |m| in Rb_VO (0.24 μ_B) indicates regions of spin density of the opposite sign, which can arise from spin polarisation of the surrounding ions whether the hole is localised or delocalised, so this quantity does not by itself fix the degree of localisation; the Löwdin analysis (Section 3.6) finds a weak antiparallel polarisation of the Ti sublattice (−0.12 μ_B in total), which accounts for essentially all of the 0.24 μ_B excess (m_{abs} − |m| is twice the integrated spin density of the minority sign).

Table 3. Converged magnetisation and Fermi energy of the production cells (U(Ti-3d) = 3.5 eV only, Γ-only). *m is the total magnetisation and m_{abs} the absolute magnetisation (Section 2.5). E_F is the Fermi energy printed by pw.x for each cell; the values are not aligned to a common reference and differences between chemically different cells are diagnostic only (Section 3.6).*

Cell	N_e	m (μ_B)	m_{abs} (μ_B)	E_F (eV, unaligned)	Lowest solution among those tested
Pristine_perfect	384	0.00	0.00	7.89	non-magnetic
Pristine_VO	378	0.00 [c]	– [c]	– [c]	non-magnetic (Section S1)
Pr_perfect	383	0.00	0.00	7.40	non-magnetic (smeared)
Pr_VO	377	0.00	0.00	9.99	non-magnetic (smeared); magnetic trial +0.28 to +0.39 eV [d]
Rb_perfect	381	1.00	1.00	7.35	one unpaired spin
Rb_VO	375	−0.97	1.21	7.13	one unpaired spin (O-2p hole, Section 3.6)
O_2 (12 Å box)	12	2.00	2.00	–	triplet

[c] Reported as non-magnetic in Section S1; the printed m_{abs} and E_F were not recorded. [d] Relative to the non-magnetic energy logged with the trial (+0.281 eV) and to the final production energy (+0.394 eV).

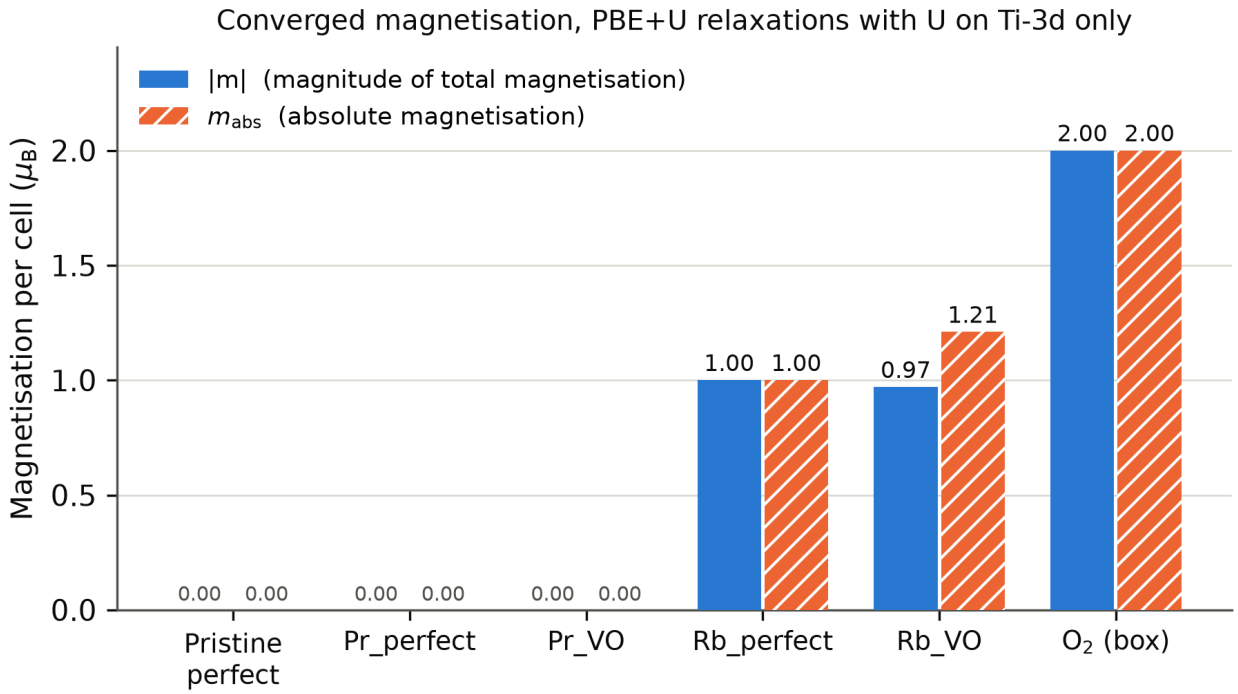


Figure 3. Magnitude of the total magnetisation (|m|) and absolute magnetisation (m_{abs}) of the production cells and of the O_2 reference (PBE+U relaxations with U on Ti-3d only). The Pr cells and the pristine cell converge to non-magnetic solutions; both Rb cells carry one unpaired spin, and Rb_VO shows $m_{abs} > |m|$.

3.4 Oxygen-site Hubbard correction: spin state of the dopant-induced holes

Because a Ti-3d-only Hubbard term provides no localising potential for holes on the oxygen sublattice, the three vacancy-free cells and Rb_VO were recomputed with an additional U = 5.5 eV on O-2p (Section 2.2), referred to below as the dual-U treatment. The pristine control is a single point; Pr_perfect and Rb_perfect were relaxed; the Rb_VO relaxation had not completed when its result was logged, so the Rb_VO entry is preliminary (Table S5). The magnetisations logged for these runs are total magnetisations m. The dual-U Rb energies were logged in eV; converted with the factor the run spreadsheet uses throughout (13.605693 eV/Ry, which reproduces the four production cells logged in both units to 0.0006 eV), they are −4324.98591 Ry (Rb_perfect) and −4284.02974 Ry (Rb_VO, incomplete relaxation), and m_{abs} was not recorded for any dual-U run (Table S5). The results are collected in Table 4. The pristine control remains non-magnetic (m = 0.00 μ_B), so the oxygen term does not induce magnetism in a cell without carriers; this control does not test whether the term biases hole-containing cells towards spin-polarised solutions. Rb_perfect changes from m = 1.00 to m = 3.00 μ_B, i.e. the three formal holes of Table 1 become three parallel unpaired spins once the oxygen term is present (their localisation has not been established from site-projected moments); Rb_VO gives m = 1.00 μ_B (preliminary), the single residual hole after two have been filled by the vacancy electrons. Pr_perfect changes from the non-magnetic smeared solution to m = 1.00 μ_B, with a maximum residual force of 0.0049 Ry/Bohr; the one formal hole of the Pr substitution thus becomes spin-polarised when the oxygen term is present. Its printed Fermi energy (6.69 eV) lies 0.53 eV below that of the dual-U pristine cell (7.22 eV), which is the direction expected when E_F is pinned at a hole state near the valence-band edge instead of lying in the gap of an insulating cell, whether or not that hole is localised, although, as for all Fermi energies in this work (Section 3.6), the unaligned difference is a qualitative indicator only. Because the localising term was placed on O-2p, the spins are expected to reside on oxygen, but this has not been confirmed from site-projected moments or spin densities. The formal hole count of each dual-U cell is matched by its spin count: one per Pr^{3+} (a trivalent state imposed by the 4f-in-core dataset, Section 2.1), three per Rb^{+} and one in Rb_VO, in line with the Kröger–Vink bookkeeping.

Table 4. Effect of an oxygen-site Hubbard term (U(O-2p) = 5.5 eV added to U(Ti-3d) = 3.5 eV) on the magnetisation of the vacancy-free and vacancy-containing cells. *m: total magnetisation; m_{abs}: absolute magnetisation (not recorded for the dual-U runs).*

Cell	m / m_{abs}, Ti-3d-only U (μ_B)	m, Ti-3d + O-2p U (μ_B)	Formal holes (Table 1)
Pristine_perfect	0.00 / 0.00	0.00 (single point)	0
Pr_perfect	0.00 / 0.00	1.00 (relaxed)	1
Pr_VO	0.00 / 0.00	not computed	−1 (net electron)
Rb_perfect	1.00 / 1.00	3.00 (relaxed)	3
Rb_VO	−0.97 / 1.21	1.00 (preliminary)	1

Two consequences follow. First, the dual-U spin count of the Rb cells equals the formal hole count and agrees with the q = +2 endpoint of the charged-cell series of Section 3.5, whereas the Ti-3d-only treatment gives one unpaired spin for the three holes of Rb_perfect. Second, the description of the Pr-substituted parent is method dependent: the hole is carried by a smeared non-magnetic solution in the Ti-3d-only treatment and by one unpaired spin, expected but not yet shown to reside on oxygen, in the dual-U treatment. The dual-U Pr_VO cell and the dual-U formation-energy cycles were not

computed, so we do not know whether the formation energies of Table 2 change under the oxygen term; the statement that formation energies are less sensitive than localisation descriptors is an expectation, not a result of this work.

3.5 Charged-cell spin series

Relaxations of Pr_VO and Rb_VO at q = +1 and +2 (uniform background, no finite-size correction) converged in 216, 549, 293 and 360 ionic steps respectively; the raw total energies are listed in Table S9. Because the periodic-image electrostatics of a charged 2×2×1 cell are uncorrected, these energies are not used to derive ionisation energies or transition levels. The spin state is used as a diagnostic. For Pr_VO both m and m_{abs} remain 0.00 μ_B at q = 0, +1 and +2: the q = +1 cell (376 electrons) has no formal carriers, and the one formal hole of the q = +2 cell (375 electrons) remains unpolarised, as in Pr_perfect under the same treatment. For Rb_VO the magnitude of the total magnetisation rises from 0.97 μ_B (q = 0) to 1.99 μ_B (q = +1) and 2.99 μ_B (q = +2); the absolute magnetisation is 1.21 μ_B at q = 0 and 3.82 μ_B at q = +2 (not recorded at q = +1), so the charged cells also contain spin density of the opposite sign. Each electron removed from the Rb cell therefore uncovers one further unpaired spin, and the q = +2 endpoint corresponds to the three holes per Rb^+ substitution found independently in the dual-U calculation of Section 3.4; the orbital character of the uncovered spins is not probed by this series. In the same Ti-3d-only treatment the neutral Rb_perfect cell, which has the same formal hole count (three) as Rb_VO at q = +2, carries only 1.00 μ_B (Section 3.3), and the one-hole Pr_VO cell at q = +2 carries none, so the number of unpaired spins per hole is not a fixed property of this treatment and the series is a count of spins rather than a test of the hole count. m_{abs} was not recorded for the q = +1 cell. The two probes are not fully independent, since both are PBE+U calculations on the same geometry family, but they rely on different mechanisms (an added localising potential versus electron removal) and agree.

3.6 Electronic-structure diagnostics

Fermi energies. The Fermi energies printed for the production cells (Table 3) are 7.89 eV (pristine), 9.99 eV (Pr_VO) and 7.13 eV (Rb_VO). They are referenced to each cell's own average electrostatic potential and are not aligned to a common vacuum or core level; in the insulating pristine cell the printed value lies in the gap at a position set by the smearing rather than by a physical level (the same Γ-only cell prints 7.89 eV in the production run and 8.24 eV in a single point of the k-mesh series at the same total energy, Table S2a). The differences (+2.10 eV for Pr_VO and −0.75 eV for Rb_VO relative to pristine; 2.86 eV between the two doped vacancy cells) should therefore be read as qualitative indicators and not as band offsets or carrier-density shifts. With that caveat, the large upward shift in Pr_VO is consistent with the net excess electron of Table 1 occupying conduction-band states in a cell containing one carrier per 570 $Å^3$ (1.75×10^{21} cm^{-3}), and the small downward shift in Rb_VO is consistent with a partly filled oxygen-derived state near the valence-band edge. At realistic ETL carrier densities (10^{17}–10^{19} cm^{-3}) the corresponding conduction-band filling would be far smaller; no estimate is attempted here.

Projected densities of states and site moments. Figure 4 shows the spin-resolved projected densities of states of Pr_VO and Rb_VO from 2×2×2 non-self-consistent calculations on the relaxed cells (Section 2.8), and Figure S2 the spin-summed densities of states. In Pr_VO the O-2p valence band ends about 2.9 eV below E_F and the Fermi level

lies in the Ti-3d conduction band. The occupied conduction-band states are 95 % Ti-3d, no feature appears inside the gap at this broadening, and the Pr-5d projection is small at both band edges. The electron count fixes 0.5 occupied states per spin channel; the broadened DOS integrates to 0.70 per channel between $E_F - 0.6$ eV and E_F because of the 0.136 eV Gaussian tails. Equal spin channels and Löwdin moments below 0.001 μ_B on every site follow from the m = 0 density on which this run is based (Section 3.3). The Pr-4f states are frozen in the core and cannot appear (Section 2.1). In Rb_VO the Fermi level lies at the top of the O-2p valence band and an empty state lies just above it in one spin channel only. In the eigenvalues it is a single minority-spin band 0.04–0.13 eV above E_F, 2.5 eV below the next minority band; in the broadened DOS it carries 0.79 states between E_F and E_F + 0.6 eV in the minority channel and none in the majority channel, 93 % of it O-2p (Ti-3d 5 %). This empty minority-spin band is the residual hole of Table 1. The Löwdin moments locate it (Figure S1, Section S2.2). Of the total of +1.01 μ_B, +0.59 μ_B sits on a single oxygen, O8, and +0.11 μ_B on each of two further oxygens, O6 and O17; the oxygen sublattice carries +1.12 μ_B in total, the Ti sublattice −0.12 μ_B (at most −0.03 μ_B on any Ti) and Rb none. O8 is bonded to the displaced Rb ion (2.56 Å) and to two Ti atoms (1.96 Å) and lies 3.39 Å from the vacant site; it is also the oxygen with the largest Bader depletion (Section 3.7). The residual hole is therefore an O-2p hole with the largest share of its Löwdin moment on one oxygen next to the dopant and the rest on neighbouring oxygens; no Ti^{3+} centre forms. Koopmans-compliant hybrid-functional calculations find that holes in bulk anatase self-trap on a single oxygen site [49,50]; the Rb_VO hole is consistent with that picture, here bound next to the dopant. Because the relaxation retained the mirror plane of the cell (Section 2.4), configurations in which the hole sits on only one oxygen of a mirror-related pair were not sampled. The sum of the magnitudes of the Löwdin moments (1.24 μ_B) is close to m_{abs} (1.21 μ_B). Löwdin moments depend on the projector basis, so these numbers locate the spin density rather than quantify it. The Kohn–Sham gap of Rb_VO from the non-self-consistent eigenvalues is 2.87 eV in the majority channel (valence-band top 0.22 eV below E_F, conduction-band minimum 2.65 eV above it); in the broadened DOS of Pr_VO the valence-band top and the conduction-band onset are 2.46 eV apart at a threshold of 0.5 states eV^{-1}, which underestimates the eigenvalue gap because of the broadening. In the run used for Figure 4 the net moment is positive, so the hole appears in the spin-down channel; the sign of m is arbitrary in a collinear calculation (the production relaxation has m = −0.97 μ_B). The Fermi energy of Pr_VO in Figure 4 (10.31 eV, 2×2×2 non-self-consistent) differs from the Γ-only self-consistent value (9.99 eV) because of the k-mesh dependence of the conduction-band filling; the Rb_VO Fermi energy is mesh-stable (7.13 versus 7.14 eV). In the non-self-consistent Rb_VO run pw.x reported 33–37 unconverged eigenvalues per k-point; in a Davidson calculation with 226 bands these are normally the highest bands, well above the hole band (band 188 of the minority channel) and the conduction-band minimum, but the upper conduction-band PDOS should be confirmed. The non-self-consistent step was not repeated with diago_full_acc = .true., and the Pr_VO eigenvalue gap was not extracted.

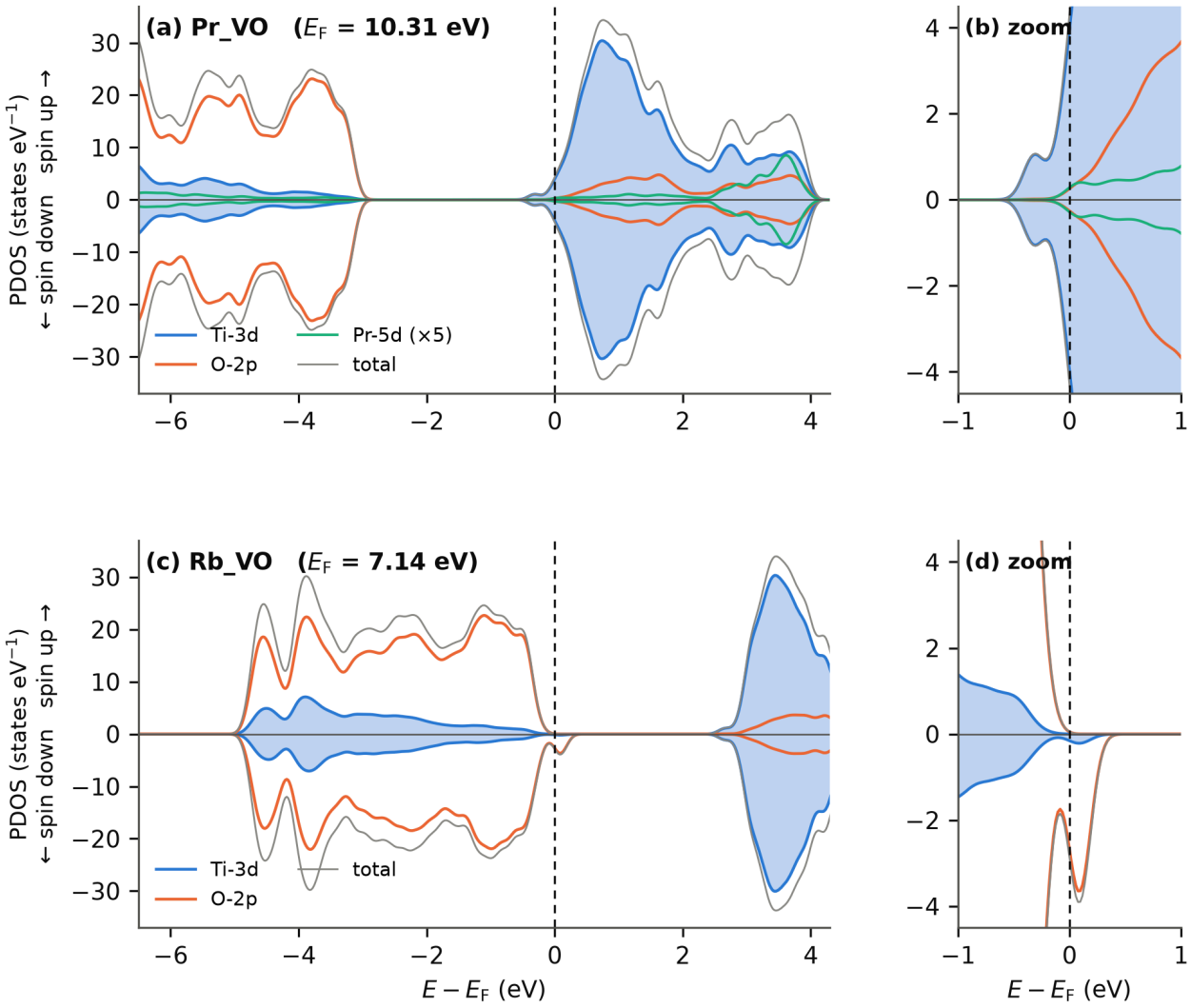


Figure 4. Spin-resolved projected densities of states of (a) Pr_VO and (c) Rb_VO, with (b, d) the region within ±1 eV of E_F, from 2×2×2 non-self-consistent calculations on the relaxed cells (PBE+U, U(Ti-3d) = 3.5 eV; tetrahedron occupations; Gaussian broadening 0.01 Ry). Spin up is plotted upwards and spin down downwards; energies are relative to each cell's Fermi level (10.31 and 7.14 eV). Filled blue: Ti-3d summed over 15 Ti; orange: O-2p summed over 31 O; green: Pr-5d multiplied by 5; grey: total. The Rb dataset has no d projector, so no dopant-d channel exists for Rb_VO. In (b) the occupied conduction-band states have equal weight in the two spin channels, as the m = 0 density implies (95 % Ti-3d; the broadened DOS holds 0.70 states per channel between −0.6 eV and E_F for 0.5 electrons). In (d) the empty band just above E_F (0.04–0.13 eV in the eigenvalues) appears in the spin-down channel only (93 % O-2p; broadened weight 0.79 states between E_F and +0.6 eV). Plotted without smoothing from the summed projwfc.x output by figures/make_pdos_figures.py.

Hybrid-functional check on the pristine cell. A single-point HSE06 calculation on the production Pristine_perfect geometry (Section 2.8, Table S5) converged to a non-magnetic solution. Its total energy and Fermi energy are not compared with the PBE+U values: the hybrid and PBE+U totals share no reference (Section 2.8), and the Fermi energies are unaligned (above). The check establishes only that the non-magnetic pristine state survives the change of functional, which is expected for a closed-shell cell without

carriers; no hybrid calculation was performed on the doped or defective cells, so the dopant contrast of Table 2 and the localisation results of Sections 3.3–3.5 have not been verified at the hybrid level.

3.7 Sensitivity tests

Hubbard U. Single-point calculations at U(Ti-3d) = 3.0, 4.0 and 4.5 eV on the production Pristine_perfect, Pr_VO and Rb_VO geometries (Table S1) leave the magnetisations unchanged in kind. Pr_VO is non-magnetic at every U, and Rb_VO carries one unpaired spin at every U (m = 1.00 μ_B in the from-scratch single points, versus m = −0.97 μ_B, m_{abs} = 1.21 μ_B in the final relaxation step at U = 3.5 eV; see Section 3.3 for the two solutions). The printed Fermi energies of Pr_VO and Rb_VO vary by at most 0.26 eV across the 1.5 eV range; that of the insulating pristine cell lies in the gap and is not a physical level (Section 3.6). The total energies vary smoothly with U, by 6.1–6.6 eV per 0.5 eV step in all three cells. The qualitative Pr/Rb distinction in spin state is therefore not tied to the particular U(Ti-3d) used at fixed geometry; formation energies were not recomputed in the sweep.

k-mesh. For Pristine_perfect, Pr_VO and Rb_VO the total energy was computed at Γ, 2×2×1, 2×2×2 and 3×3×2 meshes at U = 3.5 eV. The Γ-only energies differ from the 3×3×2 values by 1.05, 1.08 and 1.16 eV per cell (22, 23 and 25 meV per atom), whereas the 2×2×1 energies are within 1.15, 1.05 and 0.42 meV per atom of 3×3×2. The Γ-only energies lie below the 3×3×2 values, and the 2×2×2 energies of the same series lie 0.2, 1.5 and 2.1 meV per atom above them for Pr_VO, Rb_VO and Pristine_perfect (Table S2a), so convergence beyond 2×2×1 is at the 1–2 meV per atom level and is not monotonic across mesh families. The Γ-only production energies are therefore not converged in absolute terms, and the formation energies of Table 2 rely on cancellation between the perfect and defective cell of the same host rather than on convergence of each total energy. The 2×2×2 column of Table 2 quantifies the residual: −0.10 eV (Pr), −0.14 eV (Rb) and +0.04 eV on their difference. Because the k-mesh single points start from scratch, the Γ-to-2×2×2 shifts include the differences between the from-scratch and relaxation-endpoint solutions at Γ (0.038 eV for Pr_VO and 6.4 meV for Rb_VO, Table S2a). If these offsets carry over to 2×2×2, they account for essentially all of the +0.04 eV change in the contrast (leaving 0.006 eV); the perfect-cell 2×2×2 runs were not checked for such offsets. A complete formation-energy cycle exists only at Γ and, for the doped hosts, at 2×2×2. The total magnetisations are mesh-independent: 0.00 μ_B for Pristine_perfect and Pr_VO and 0.99–1.00 μ_B for Rb_VO at every mesh, the latter being the from-scratch Rb_VO solution of Section 3.3.

Dispersion. D3(BJ) single points on the relaxed Pr_VO and Rb_VO cells gave dispersion contributions of −10.29 and −10.09 eV (Table S5). Because D3 is an additive geometric correction, the electronic part of each single point should reproduce the production total energy. It does so to 0.006 eV for Rb_VO but lies 0.038 eV above the production value for Pr_VO. The electronic part of the Pr_VO D3 single point is identical, to 10^{-5} Ry, to the from-scratch Γ single point of the k-mesh series (Table S2a), so the offset is common to the from-scratch single points and is not caused by the D3 term; its origin (a different smeared solution, a smearing difference or the manually stopped relaxation) has not been established. The dispersion energies of the perfect substituted cells and of the O_2 reference were not computed, so the dispersion correction to $E_f(V_O)$ and to ΔE_f cannot be reconstructed from these results: the correction to the contrast requires [D3(Pr_VO) − D3(Pr_perfect)] − [D3(Rb_VO) − D3(Rb_perfect)], of which only the two defective-cell

terms are available. The effect of dispersion on the vacancy energetics is therefore undetermined. The D3 single points give m = 0.00 μ_B (Pr_VO) and 1.00 μ_B (Rb_VO) and Fermi energies within 0.02 eV of the production values; the Rb_VO value is the from-scratch solution discussed in Section 3.3, not the production solution (m = −0.97 μ_B, m_{abs} = 1.21 μ_B).

Alternative vacancy sites in the Pr cell. Three further Pr_VO cells were relaxed with the vacancy at oxygen sites 1, 7 and 17 of the orchestration input (Table S4; these labels are not the pw.x atom numbers used elsewhere). Sites 1 and 17 are symmetry-equivalent second-shell positions (Pr–O = 3.99 Å) and give E_f = +2.07 eV each, agreeing to 0.1 meV; the more distant site 7 (4.94 Å) gives +2.72 eV. The production site is thus the lowest of the four sampled sites by 0.78 eV. This is a comparison among selected configurations in a 2×2×1 cell, where “distant” sites are still within one lattice parameter of the periodic images of the dopant. It indicates a preference for the production site over the two more distant positions sampled but is not a global search. Read as a dopant–vacancy association energy in this cell, moving the vacancy from the most distant sampled site to the production site releases 1.43 eV (0.78 eV from the second shell); because the distant site lies within one lattice parameter of periodic images of the dopant, this is not a dilute-limit binding energy. The production site is a first-shell oxygen of the dopant site in the Rb cell, and the Löwdin populations of Pr_VO are consistent with the same site (Section 2.3); other first-shell sites were not computed, site multiplicities are not given, and no corresponding sampling was performed for the Rb cell. The positions of the removed oxygens are listed in Table S4.

Bader charges. Bader analysis of the relaxed Pr_VO and Rb_VO cells (Table S3) places 8.96 e on Pr and 8.13 e on Rb, i.e. charges of +2.04 e and +0.87 e relative to the PAW valence (±0.03 e; values implied by the recorded electron totals and sublattice means, Table S3), consistent with the +3 and +1 formal states after the usual reduction of Bader charges relative to formal values. The per-atom populations were not re-extracted for this revision, so the dopant values are those implied by the recorded totals and sublattice means; grid convergence of the Bader charges was not tested (Section 2.8). Relative to the Pr_VO cell, the 31 oxygen atoms of Rb_VO hold 0.97 e less in total, with the depletion concentrated on one oxygen bonded to the displaced Rb ion (−0.26 e, O8; Rb–O8 = 2.56 Å, 3.39 Å from the vacant site) and a second oxygen (−0.14 e, O30; 3.07 Å from Rb), with five further oxygens contributing −0.05 to −0.09 e each. The 15 Ti atoms show a uniform mean depletion of −0.012 e (−0.18 e in total) and no Ti differs by more than 0.05 e. No Ti site shows the excess of a few tenths of an electron, relative to the other Ti sites, that a localised Ti^{3+} centre would produce, although the expected size of that signal was not calibrated on a reference cell containing a Ti^{3+} centre, so the test is indicative rather than decisive. The comparison is between cells of different composition, so the differences contain chemical polarisation by the dopant as well as carrier redistribution, and Bader charges are not orbital occupations; the two cells also differ by two carriers (the net excess electron of Pr_VO and the residual hole of Rb_VO, Table 1), whose contributions to the oxygen depletion this comparison cannot separate; the result is therefore compatible with an oxygen-centred hole in Rb_VO, but it is not a hole count and carries no spin information. The Löwdin analysis of Section 3.6 supplies the spin information: the oxygen with the largest Bader depletion, O8, also carries the largest spin moment (+0.59 μ_B). A same-composition comparison (Rb_VO at q = 0 versus q = +1, or the spin density itself) would be more direct.

Uncertainty of the main results. Of the numerical settings, only the k-mesh was varied for the formation energies: it shifts the individual values by 0.10–0.14 eV and the Pr–Rb contrast by 0.04 eV, an amount that the from-scratch offsets of the 2×2×2 single points could account for. The plane-wave cut-off, the smearing entropy, the residual forces, the supercell size and the oxygen-reference error were not quantified (Section 5). The O_2 reference error and any error common to the two doped hosts cancel in the contrast; the untested contributions would have to amount to several eV to change the sign of the Rb value (−4.06 eV) or to remove the contrast (5.36 eV), whereas the sign of the Pr value (+1.29 eV) and the reductions relative to the pristine host are more sensitive, the latter also to the relaxation status of the pristine perfect cell (Section 2.6).

4. Discussion

4.1 The energetic ordering follows the formal electron count. The three formation energies of Table 2 fall in the order pristine (+4.74 eV) > Pr (+1.29 eV) > Rb (−4.06 eV), which is the order of the formal hole count of the vacancy-free cells (0, 1, 3; Table 1) and also of increasing cation radius (Section 4.3). A neutral vacancy releases two electrons. In the pristine cell both must occupy conduction-band-derived states, and the full cost of oxygen removal is paid. In the Pr cell one of the two can instead fill the single acceptor-like hole introduced by Pr^{3+}, and in the Rb cell both can fill two of the three holes introduced by Rb^{+}. Transferring an electron from a conduction-band-like state to an oxygen-2p-derived hole near the valence-band edge recovers an energy of the order of the band gap, so reductions of a few eV per transferred electron are expected; the reductions found (at least 3.45 eV for one electron in the Pr cell and 8.81 eV for two in the Rb cell; lower bounds because the pristine reference cell is unrelaxed, Section 2.6) are of that order but not equal per electron. The Kohn–Sham gap of the Rb_VO cell is 2.87 eV and the band separation in the broadened densities of states of Pr_VO about 2.5 eV (Section 3.6), both smaller than either per-electron reduction (3.45 eV for Pr, 4.40 eV per electron for Rb), so electron transfer across the gap does not account for the reductions on its own. The remainder would have to come from other contributions, for example the electrostatic interaction between the vacancy and the dopant, which the dependence of E_f on the vacancy site in the Pr cell indicates (Section 3.7), lattice relaxation (the Rb ion lies 1.27 Å off its ideal site in Rb_VO, Section 3.1), band filling and interactions with periodic images; the two-dopant design cannot decompose the formation-energy difference into these contributions.

4.2 Relation to previous work. The Rb result parallels, in anatase, the behaviour that Carey and Nolan found for alkali dopants in α-Cr_2O_3, where the V_O formation energy is strongly reduced and oxygen holes are localised with an O-2p Hubbard term [16]. It also agrees in sign with the experimental vacancy-formation trend reported for Rb-doped TiO_2 by Tian et al. [15]. Because only one alkali was studied here, the radius dependence found by Carey and Nolan across Li–Rb is not tested. The Pr result is consistent with the trivalent-dopant picture of Iwaszuk and Nolan for rutile [17]: the vacancy is not spontaneous at the oxygen-rich reference, but its cost is reduced by at least 3.45 eV relative to the pristine host, so vacancy formation near Pr becomes accessible as the oxygen chemical potential is lowered; one vacancy per Pr over-compensates the dopant (Table 1), and the fully compensated $2Pr_{Ti} + V_O$ configuration that corresponds to vacancy compensation of a trivalent dopant was not computed. With $E_f(V_O; Pr) = 1.29$ eV $+ \Delta\mu_O$, oxygen removal near Pr has a negative formation energy once μ_O falls more than 1.29 eV below ½E(O_2), a threshold that carries the common offset of the PBE O_2

reference (Section 2.7); whether that lies within the stability field of TiO_2 against lower oxides was not evaluated (Section 5). For Rb the formation energy is negative already at the oxygen-rich limit and decreases further as μ_O is lowered, so a vacancy-free Rb_{Ti} cell is not a stable reference at any oxygen chemical potential at which TiO_2 exists; whether Rb substitutes on the Ti site at all, rather than occupying interstitial or surface sites or forming a secondary phase, was not evaluated. The spin polarisation of the Pr-induced hole when an oxygen-site U is present parallels the oxygen-hole states reported for trivalent dopants in rutile by Iwaszuk and Nolan [17].

4.3 What the two-dopant comparison does and does not isolate. Pr^{3+} and Rb^{+} differ in formal charge (−1 versus −3 e relative to Ti^{4+}), in ionic radius (0.99 versus 1.52 Å [26]), in bonding (a 5d-bearing trivalent cation versus a closed-shell alkali) and in accessible oxidation states. Both are larger than Ti^{4+} (0.605 Å), so the comparison does show that a large size mismatch is compatible with either sign of $E_f(V_O)$ at the oxygen-rich reference; it does not hold size constant, and no strain or fixed-geometry decomposition was computed, so the contrast cannot be apportioned between electronic and elastic contributions. What can be said is that the sign of the contrast and the size of the reductions are consistent with the electron count; because the ordering pristine > Pr > Rb is also the order of increasing cation radius (0.605, 0.99 and 1.52 Å) and Carey and Nolan found a radius dependence at fixed alkali valence [16], a size-based explanation of the ordering is not excluded. The relaxed Rb_VO cell shows that an elastic channel exists: the Rb ion lies 1.27 Å off its ideal site, in the space vacated by the adjacent oxygen (Section 3.1), but the energy of this relaxation has not been separated from the electronic contribution, and the corresponding Pr geometry has not been analysed. Isolating the two variables would require a controlled series within the same protocol, for example La^{3+}, Y^{3+} and Sc^{3+} (same valence, radii from 1.03 to 0.745 Å [26]) and K^{+} and Na^{+} (same valence as Rb^{+}, smaller radii), together with fixed-geometry single points to separate electronic from relaxation contributions.

4.4 The Pr-substituted parent is method dependent. The vacancy-free Pr cell is non-magnetic with U on Ti-3d only and carries one unpaired spin (m = 1.00 μ_B), expected but not yet shown to reside on oxygen, with U on O-2p (Section 3.4). Both solutions contain the one formal hole; the dual-U result is the one in which the spin count equals the hole count, as in the dual-U Rb cells, but it rests on a U(O-2p) value transferred from Cr_2O_3 and on calculations that have not been extended to the vacancy cell or to the formation-energy cycle. Until those are available, the Pr-substituted parent should be described as having one hole whose localisation depends on the treatment of oxygen correlation; it is neither closed-shell nor demonstrably trap-free. For the vacancy-containing Pr cell, the Ti-3d-only treatment places the net carrier, one excess electron, in conduction-band-derived states (Table 1 for the count, Figure 4 for the placement); this conclusion rests on a single smeared non-magnetic solution and one magnetic trial, has not been tested with the oxygen-site term, and concerns a cell with one carrier per 570 $Å^3$.

4.5 Implications for dopant screening in oxide electron-transport layers. The results are consistent, within this protocol and host, with the formal charge mismatch as a candidate screening variable for how strongly a substitutional acceptor lowers the oxygen-vacancy formation energy: both dopants studied lower it, the trivalent dopant leaving the vacancy uphill at the oxygen-rich limit and the monovalent dopant making it spontaneous there. Extending this to other dopants of the same formal charge, and the three statements that follow, are hypotheses that this work motivates but does not test.

(i) Trivalent rare-earth dopants other than Pr (La, Nd, Sm) should behave like Pr in the V_O energetics, because they share the −1 e mismatch. (ii) Pentavalent donors (Nb^{5+}, Ta^{5+}) should raise rather than lower $E_f(V_O)$, having a +1 e mismatch, the electronic counterpart of the trivalent case. (iii) The acceptor states introduced by alkali substitution are candidates for recombination centres at an ETL/absorber interface. Testing (iii) requires interface models, charge-transition levels with finite-size corrections and carrier-capture calculations, none of which are reported here, so no ranking of dopants by device performance is made.

5. Limitations

i. *Supercell and concentration.* One dopant and one vacancy per 2×2×1 cell correspond to 6.25 % cation substitution and 3.1 % anion vacancies. Defect–image interactions and band filling by the net carrier of each vacancy cell are not corrected; absolute formation energies at the dilute limit may differ by several tenths of an eV, and because the Pr cell carries a conduction electron and the Rb cell a valence hole these terms need not cancel in ΔE_f. (ii) *Brillouin-zone sampling.* Production relaxations are Γ-only. Γ-only total energies lie 22–25 meV per atom below the 3×3×2 values (Section 3.7); formation energies rely on cancellation within each host, and the 2×2×2 check shifts each doped-host E_f by 0.10–0.14 eV and ΔE_f by 0.04 eV, which estimates rather than bounds the residual; the pristine E_f was not checked. (iii) *Plane-wave cut-off.* 40 Ry was not tested against a higher cut-off; a comparable anatase study uses 55 Ry [13]. (iv) *Force tolerance.* 0.005 Ry/Bohr (0.129 eV/Å) is loose; the Pr_VO relaxation was stopped manually and the pristine V_O relaxation ended at 0.0056 Ry/Bohr. The production pristine perfect cell was not relaxed (residual force 0.0225 Ry/Bohr, Section 2.6), so the pristine formation energy and the two reductions measured from it are lower bounds, short by its relaxation energy, which was not computed. The effect of tighter relaxation on the energetics and on the localisation results has not been tested. (v) *Spin search.* The production relaxations of the doped cells started from zero magnetisation (the pristine V_O cell from 0.05; Section 2.5), and one magnetic trial was performed, for Pr_VO; occupations, smearing and symmetry settings were not varied, and two spin solutions of Rb_VO were found that differ by 6.4 meV (Section 3.3). Because E_f is a difference of total energies, a lower-energy spin solution of any parent or vacancy cell, in particular of Pr_perfect, which was not given a magnetic trial, would shift the corresponding E_f by the same amount. (vi) *Pr-4f treatment.* The Pr dataset freezes the 4f electrons in the core. Mechanisms involving 4f occupation, 4f hybridisation or a Pr^{3+}/Pr^{4+} change of oxidation state cannot be examined with it, and none is claimed; the trivalent state, and with it the formal carrier counts of the Pr cells in Table 1, is imposed by this choice rather than tested. (vii) *Oxygen-site U.* U(O-2p) = 5.5 eV is transferred, not derived for anatase; the dual-U treatment was applied to the vacancy-free cells and Rb_VO (whose relaxation was not completed when logged) but not to Pr_VO or to the formation-energy cycles. (viii) *Charged cells.* No electrostatic or potential-alignment correction was applied; charged-cell energies are not interpreted. An ion-clamped dielectric constant of $\varepsilon_\infty \approx 5.8$ (experimental in-plane value, 5.82, as tabulated by Arrigoni and Madsen [14]) would give a first-order monopole correction [39] of a few tenths of an eV for q = +1 at this cell size and four times that for q = +2. The static constant of anatase is much larger and would reduce the estimate, so these numbers are upper bounds and are not used. (ix) *Oxygen reference.* All formation energies are at $\mu_O = ½E(O_2)$ (oxygen-

rich limit) without finite-temperature corrections; equilibrium vacancy concentrations, dopant incorporation energies, the stability field of TiO_2 against lower oxides and competing Pr and Rb oxides were not evaluated, and the O_2 overbinding of PBE was not corrected; no statement about the synthesisable phase of Rb-doped anatase is therefore made. (x) *Defect channels and stoichiometry.* Only a single V_O per cell was computed; sequential vacancies, the fully compensated 2 Pr + 1 V_O and 2 Rb + 3 V_O stoichiometries, cation vacancies, interstitials (including Rb interstitials, a plausible alternative for a large monovalent ion), surface sites and dopant clustering were not. (xi) *Functional.* The HSE06 check concerns the pristine cell only; the doped and defective cells have not been verified with a hybrid functional. (xii) *Dispersion.* The D3 correction to the formation energies is undetermined because the parent-cell terms were not computed. (xiii) *Charge and spin analysis.* The Bader comparison is between cells that differ in composition and by two carriers. The Löwdin moments that place the Rb_VO hole mainly on one oxygen depend on the projector basis and have not been cross-checked with a spin-density isosurface or another partitioning, and the relaxations retained the mirror plane of the defect cell, so hole configurations that break it were not sampled; for Pr_perfect and the dual-U cells no site-resolved analysis exists. (xiv) *Scope.* No surface, interface, transition-level, capture-coefficient or device calculation is reported; statements about electron-transport layers are hypotheses.

6. Conclusions

Using one PBE+U protocol in the same 2×2×1 anatase supercell, we find neutral oxygen-vacancy formation energies of +4.74 eV (pristine; a lower bound, because the perfect reference cell was not relaxed), +1.29 eV (Pr-substituted) and −4.06 eV (Rb-substituted) at the oxygen-rich reference. Both substitutions lower the cost of oxygen removal; Pr leaves it uphill and Rb makes it spontaneous. The Pr–Rb contrast of 5.36 eV changes by 0.04 eV between Γ-only and 2×2×2 sampling. The ordering follows the formal electron count: one hole per Pr^{3+} and three per Rb^{+}, with a single vacancy over-compensating Pr and leaving one residual hole in the Rb cell. The Rb cells carry unpaired spin in every treatment tested. In Rb_VO the residual hole is an empty minority-spin O-2p band whose Löwdin moment sits mainly on one oxygen bonded to the Rb ion (0.59 of 1.01 μ_B), and the Rb ion lies 1.27 Å off its ideal site towards the vacancy; a charged-cell series reaches three unpaired spins per Rb^{+}. The Pr cells converge to non-magnetic smeared solutions with U on Ti-3d only, but the vacancy-free Pr cell develops an unpaired spin when U is also placed on O-2p, so the spin state of the hole in Pr-substituted anatase is method dependent and a closed-shell, trap-free description of that material is not supported. The two-dopant comparison is consistent with charge-compensation control of oxygen-vacancy energetics in this host and provides common-protocol values for one trivalent and one monovalent dopant, but it does not separate oxidation state from ionic size and bonding, and it does not establish interfacial recombination behaviour. Completing the dual-U formation-energy cycles, tightening the relaxations and adding a matched-valence dopant series are the natural next steps.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to restructure and edit the text, to check the internal consistency of numbers, cross-references and citations, and to write the Python scripts that plot Figures 1–4, S1 and S2 and the graphical abstract from the authors' Quantum ESPRESSO outputs and computed values; no image was generated or altered by the tool. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. All calculations, analyses, interpretations and conclusions are the authors' own.

CRediT authorship contribution statement

Mohammad Ghadimimehr: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Azimah Binti Omar:** Supervision, Methodology, Validation, Writing – review & editing. **Siti Rohani Binti Sheikh Raihan:** Resources, Writing – review & editing.

Data availability

The Quantum ESPRESSO input files, converged output files, pseudopotential files, Bader output tables, projected-density-of-states and Löwdin data, the full-precision energy ledger (Table S9), the run manifest (Table S10) and the Python scripts that generate every figure are available from the corresponding author on request and will be deposited in a public repository (Zenodo) with a citable DOI on acceptance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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